Data Warehouse

Information Technology Institute

Gravity Books Analytics

Project Overview

Purpose of the Project

This project implements a Sales Data Warehouse designed to transform raw transactional sales data into structured, analytics-ready datasets. The system consolidates customer, product, and transaction data into a dimensional model to support reporting and business intelligence.

Business Problem it Solves

Operational systems are optimized for transactions, not analytics. Reporting directly from raw transactional data leads to:

- Slow query performance
- Inconsistent calculations
- Data duplication
- Lack of historical tracking

This project solves these issues by:

- Cleaning and standardizing raw sales data
- Creating structured dimension tables
- Centralizing business logic (e.g., revenue calculation)
- Providing a single source of truth for reporting

Type of Data Being Processed

- Sales transactions
- Customer master data
- Product master data
- Pricing and discount information
- Currency information

The data includes structured transactional records and reference data.

Who Uses the Data

The data warehouse is used by:

- Data analysts
- Business intelligence developers
- Dashboard/report developers
- Finance teams
- Sales management

System Architecture

Data Sources

The OLTP system contains:

Core Transaction Tables

- customer
- cust_order
- order_line
- order_history
- order_status
- shipping_method

Product-Related Tables

- book
- author
- book_author (bridge)
- publisher
- book_language

Customer Location Tables

- address
- country
- customer_address
- address_status

Ingestion method

- Batch ETL using SSIS
- Daily incremental loads
- Dimension load before fact load

Processing tools/frameworks

- ETL pipelines (e.g., SSIS or equivalent orchestration tool)
- SQL-based transformation logic

Storage Layers

Layer 1 – Source (OLTP)

Normalized transactional schema.

Layer 2 – Staging

Temporary transformation tables.

Layer 3 – Data Warehouse (Star Schema)

Final Star Schema Tables

Fact Table

- Sales_fact

Dimension Tables

- Customer_dim
- Book_dim
- Author_dim
- Bridge_Book_Author
- Address_dim
- ShippingMethod_dim
- OrderStatus_dim
- Date_dim

Data Flow Explanation

1. Extract data from OLTP system.
2. Load staging tables.
3. Transform:
   - Flatten normalized structures.
   - Generate surrogate keys.
   - Apply SCD logic (Customer_dim, Address_dim).
4. Load dimensions.
5. Load Sales_fact with foreign keys.
6. BI tools query the star schema only.

Data Pipelines Documentation

Pipeline 1: Customer Dimension Load

Source:

- customer

Target:

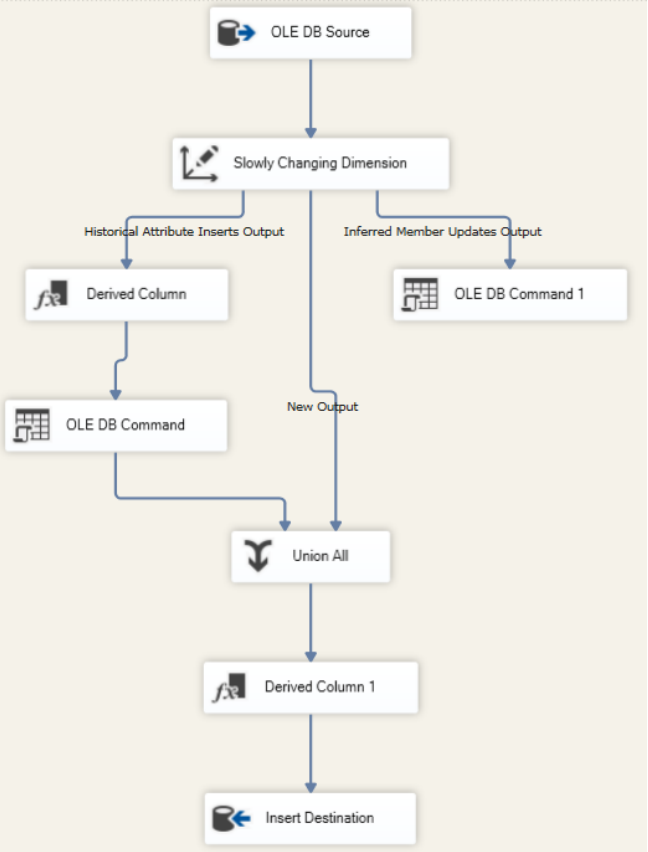
- Customer_dim

Type:

- SCD Type 2

Logic

- Track changes in:
  - first_name
  - last_name
  - email
- Maintain:
  - effective_start_date
  - effective_end_date
  - is_active



Pipeline 2: Address Dimension Load

Source:

- address
- country
- customer_address
- address_status

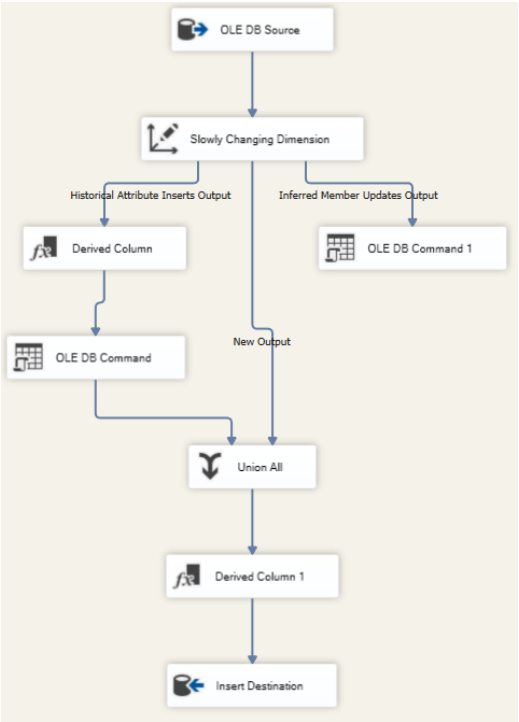
Target: Address_dim

Type:

- SCD Type 2

Logic

- Join address + country
- Track address changes historically
- Manage active flag



Pipeline 3: Book Dimension Load

Source:

- book
- publisher
- book_language

Target:

- Book_dim

Logic

- Flatten publisher_name
- Flatten language_name
- No SCD (assumed Type 1)

Pipeline 4: Author Dimension Load

Source:

- author

Target:

- Author_dim

Simple dimension with surrogate key.

Pipeline 5: Bridge Book Author Load

Source:

- book_author

Target:

- Bridge_Book_Author

Resolves many-to-many relationship between books and authors.

Pipeline 6: Shipping Method Dimension

Source:

- shipping_method

Target: ShippingMethod_dim

Pipeline 7: Order Status Dimension

Source:

- order_status

Target:

- OrderStatus_dim

Pipeline 8: Date Dimension

Source: Generated calendar table

Contains:

- day
- month
- month_name
- quarter
- year
- weekday
- is_weekend

Pipeline 9: Sales Fact Load

Source:

- cust_order
- order_line
- order_history

Target: Sales_fact

Grain

- One record per order line per order.

Foreign Keys

- date_key
- customer_key
- book_key
- shipping_method_key

- shipping_address_key
- status_key

Measures

- quantity
- unit_price
- sales_amount

Data Schema and Data Catalog

Fact Table

Table: Sales_fact

Table Type: Fact Table

Grain: One record per order line per book id.

Column Name	Data Type	Key Type	Business Purpose
sales_key	INT	Surrogate Key (PK)	Unique warehouse identifier for each fact row.
date_key	INT	Foreign Key	Links sale to Date_dim
customer_key	INT	Foreign Key	Links sale to Customer_dim
book_key	INT	Foreign Key	Links sale to Book_dim
shipping_method_key	INT	Foreign Key	Links to ShippingMethod_dim
shipping_address_key	INT	Foreign Key	Links to Address_dim
status_key	INT	Foreign Key	Links to OrderStatus_dim
order_id	INT	Business Key (Degenerate)	Original order identifier from source
quantity	INT	Measure	Number of books sold
unit_price	DECIMAL(10,2)	Measure	Selling price per unit
sales_amount	DECIMAL(12,2)	Measure	Total revenue (quantity × unit_price)

Dimensions Tables

Table: Customer_dim

Table Type: Dimension

SCD Type: Type 2

Column Name	Data Type	Key Type	Business Purpose
customer_key	INT	Surrogate Key (PK)	Warehouse identifier
customer_id	INT	Business Key	Source system identifier
first_name	VARCHAR(200)	Attribute	Customer first name
last_name	VARCHAR(200)	Attribute	Customer last name
email	VARCHAR(350)	Attribute	Customer email
effective_start_date	DATETIME2	SCD Field	Record validity start date
effective_end_date	DATETIME2	SCD Field	Record validity end date
is_active	BIT	SCD Field	Indicates active record

Table: Address_dim

Table Type: Dimension

SCD Type: Type 2

Column Name	Data Type	Key Type	Business Purpose
address_key	INT	Surrogate Key (PK)	Warehouse identifier
address_id	INT	Business Key	Source address ID
street_number	VARCHAR(10)	Attribute	Street number
street_name	VARCHAR(200)	Attribute	Street name
city	VARCHAR(100)	Attribute	City name
country_name	VARCHAR(200)	Attribute	Country name
address_status	VARCHAR(30)	Attribute	Address status
effective_start_date	DATETIME2	SCD Field	Record validity start
effective_end_date	DATETIME2	SCD Field	Record validity end
is_active	BIT	SCD Field	Indicates active version

Table: Book_dim

Table Type: Dimension

Column Name	Data Type	Key Type	Business Purpose
book_key	INT	Surrogate Key (PK)	identifier
book_id	INT	Business Key	Source book ID
title	VARCHAR(400)	Attribute	Book title
isbn13	VARCHAR(13)	Attribute	ISBN identifier
language_name	VARCHAR(50)	Attribute	Book language
num_pages	INT	Attribute	Number of pages
publication_date	DATE	Attribute	Publication date
publisher_name	NVARCHAR(1000)	Attribute	Publisher name

Table: Author_dim

Table Type: Dimension

Column Name	Data Type	Key Type	Business Purpose
author_key	INT	Surrogate Key (PK)	Warehouse author identifier
author_id	INT	Business Key	Source author ID
author_name	VARCHAR(400)	Attribute	Full author name

Table: Bridge_Book_Author

Table Type: Dimension

Column Name	Data Type	Key Type	Business Purpose
book_key	INT	Foreign Key	Links to Book_dim
author_key	INT	Foreign Key	Links to Author_dim

Table: ShippingMethod_dim

Table Type: Dimension

Column Name	Data Type	Key Type	Business Purpose
method_key	INT	Surrogate Key (PK)	Warehouse shipping method ID
method_name	VARCHAR(100)	Attribute	Shipping method name
cost	DECIMAL(6,2)	Attribute	Shipping cost

Table: OrderStatus_dim

Table Type: Dimension

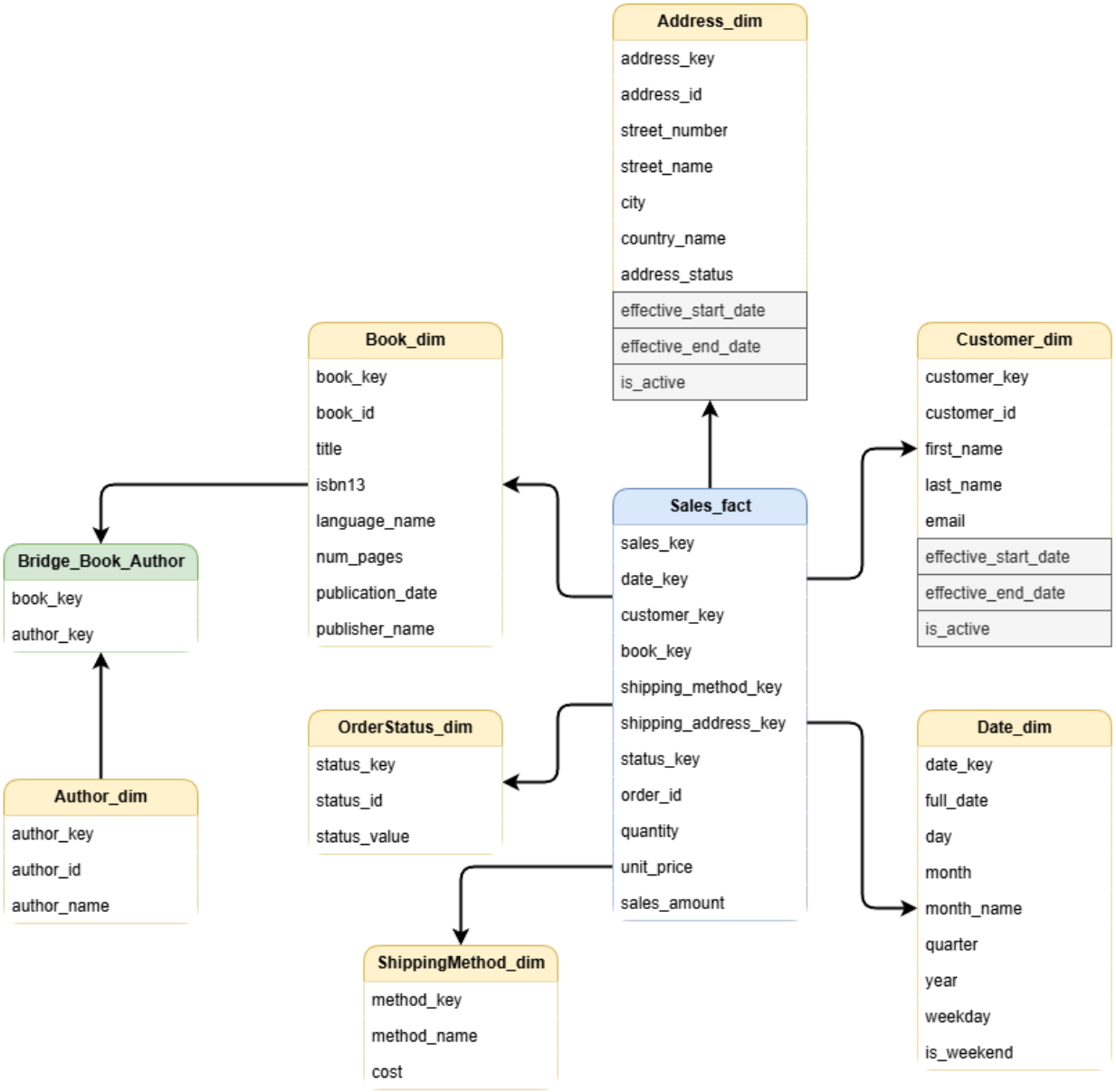
Column Name	Data Type	Key Type	Business Purpose
status_key	INT	Surrogate Key (PK)	identifier
status_id	INT	Business Key	Source status ID
status_value	VARCHAR(20)	Attribute	Status description

Table: Date_dim

Table Type: Dimension

Column Name	Data Type	Key Type	Business Purpose
date_key	INT	Surrogate Key (PK)	Date in YYYYMMDD format
full_date	DATE	Attribute	Actual calendar date
day	INT	Attribute	Day of month
month	INT	Attribute	Month number
month_name	VARCHAR(20)	Attribute	Month name
quarter	INT	Attribute	Quarter number
year	INT	Attribute	Year
weekday	VARCHAR(20)	Attribute	Day name
is_weekend	BIT	Attribute	Weekend indicator

Data Modeling



Business Rules

Sales Amount Calculation

- sales_amount = quantity × unit_price

Status Logic

- Latest status from order_history determines fact status.

Address Logic

- Shipping address used from cust_order.dest_address_id.

SCD Rules

- Customer and Address use Type 2.

Data Quality Rules

- No null foreign keys in fact table
- Quantity > 0
- Price ≥ 0
- Valid status values
- No duplicate order_line entries
- Referential integrity enforced

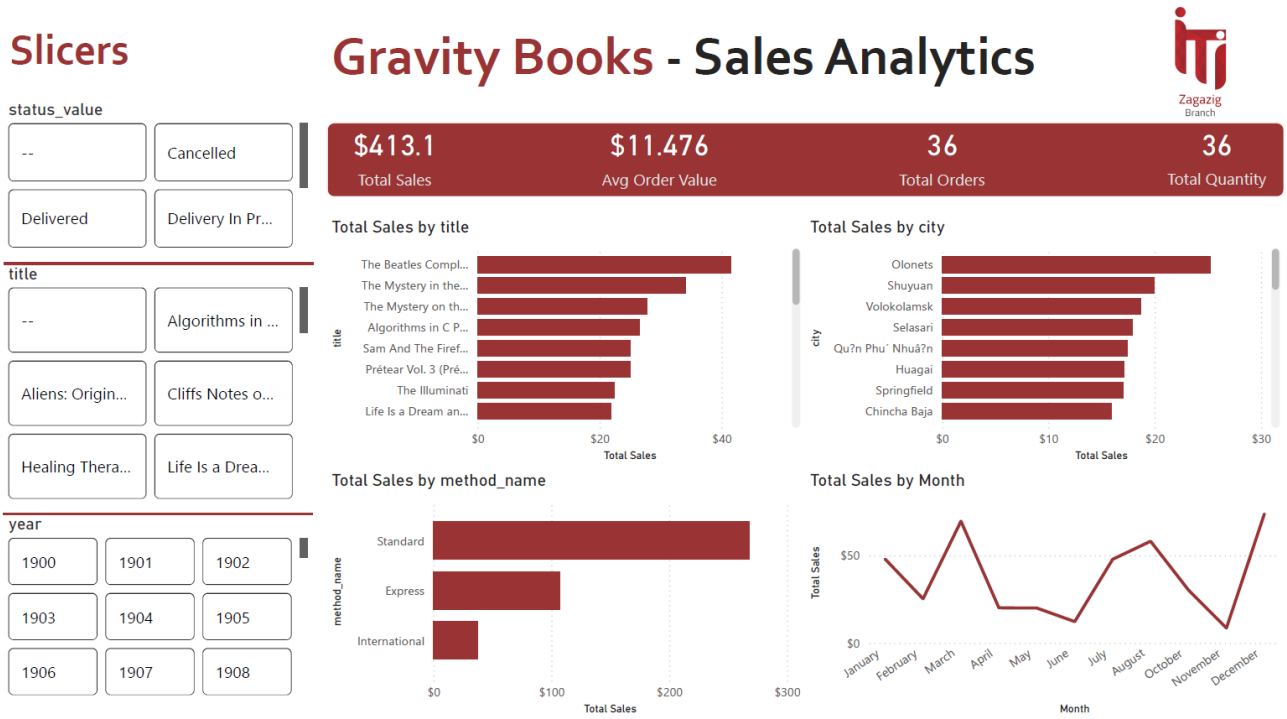
Failed rows: Redirected to error table

Data Ownership

Business Owner

- Head of Sales Operations

Technical Owner

- Data Engineering Team

Analytics using Power BI